33.3: Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms

An evolutionary tree highlights lophotrochozoa, which is the first branch for bilateria.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia. Lophotrochozoa, Ecdysozoa, and Deuterostomia are collectively labeled Bilateria.

The vast majority of animal species belong to the clade Bilateria, whose members exhibit bilateral symmetry and are triploblastic (see Concept 32.3). Most bilaterians also have a digestive tract with two openings (a mouth and an anus) and a body cavity (a coelom or hemocoel; see Figure 32.9). Recent DNA analyses suggest that the common ancestor of living bilaterians lived about 670millionyears ago. To date, however, the oldest fossil that is widely accepted as a bilaterian is of *Kimberella*, a mollusc (or close relative) that lived 560millionyears ago (see Figure 32.5b). Many other bilaterian groups first appeared in the fossil record during the Cambrian explosion (535 to 525 million years ago).

Molecular evidence suggests that today there are three major clades of bilaterally symmetrical animals: Lophotrochozoa, Ecdysozoa, and Deuterostomia. This section will focus on the first of these clades, the lophotrochozoans. Concepts 33.4 and 33.5 will explore the other two clades.

Although the clade Lophotrochozoa was identified by molecular data, its name comes from features found in some of its members. Some lophotrochozoans develop a structure called a *lophophore*, a crown of ciliated tentacles that functions in feeding, while others go through a distinctive stage called the *trochophore larva* (see Figure 32.12). Other members of the group have neither of these features. Few other unique morphological features are widely shared within the group—in fact, the lophotrochozoans are the most diverse bilaterian clade in terms of body plan. This diversity in form is reflected in the number of phyla classified in the group: Lophotrochozoa includes 18 phyla, more than twice the number in any other clade of bilaterians.

We'll now introduce six of the diverse lophotrochozoan phyla: the flatworms, rotifers and acanthocephalans, ectoprocts, brachiopods, molluscs, and annelids.

Flatworms

Flatworms (phylum Platyhelminthes) live in marine, freshwater, and damp terrestrial habitats. In addition to free-living species, flatworms include many parasitic species, such as flukes and tapeworms. Flatworms are so named because they have thin bodies that are flattened dorsoventrally (between the dorsal and ventral surfaces); the word *platyhelminth* means “flat worm.” (Note that *worm* is not a formal scientific name but rather refers to animals with long, thin bodies.) The smallest flatworms are nearly microscopic free-living species, while some tapeworms are more than 20 m long.

Although flatworms are triploblastic, they lack a body cavity. Their flat shape increases their surface area, placing all their cells close to water in the surrounding environment or in their gut. Because of this proximity to water, gas exchange and the elimination of nitrogenous waste (ammonia) can occur by diffusion across the body surface. As shown in **Figure 33.8**, a flat shape is one of several structural features that maximize surface area and have arisen (by convergent evolution) in different groups of animals and other organisms.

Figure 33.8

AP Make Connections: Maximizing Surface Area

Make connections: Maximizing surface area.

The diagram reads as follows.

• Maximizing surface area: In general, the amount of metabolic or chemical activity an organism can carry out is proportional to its mass or volume. Maximizing metabolic rate, however, requires the efficient uptake of energy and raw materials, such as nutrients and oxygen, as well as the effective disposal of waste products. For large cells, plants, and animals, these exchange processes have the potential to be limiting due to simple geometry. When a cell or organism grows without changing shape, its volume increases more rapidly than its surface area (see figure 6.7). As a result, there is proportionately less surface area over which exchange processes can occur. The challenge posed by the relationship of surface area and volume occurs in diverse contexts and organisms, but the evolutionary adaptations that meet this challenge are similar. Structures that maximize surface area through flattening, folding, branching, and projections have an essential role in biological systems.

• Flattening: By having a body that is only a few cells thick, an organism such as a flatworm can use its entire body surface for exchange. (See figure 40.3).

• Folding: A TEM shows portions of two chloroplasts in a plant leaf. Photosynthesis occurs in chloroplasts, which have a flattened and interconnected set of internal membranes called thylakoid membranes. The

foldings of the thylakoid membranes increase their surface area, enhancing the exposure to light and thus increasing the rate of photosynthesis. (See figure 10.4).

• Branching: Water uptake relies on passive diffusion. The highly branched filaments of a fungal mycelium increase the surface area across which water and minerals can be absorbed from the environment. (See figure 31.2).

• Projections: In vertebrates, the small intestine is lined with finger-like projections called villi that absorb nutrients released by the digestion of food. Each of the villi shown here is covered with large numbers of microscopic projections called microvilli, resulting in a total surface area of about 300 m^2 in humans, as large as a tennis court. (See figure 41.12).

Diagram compares surface area to volume for two different shapes with the same volume. Both shapes are subdivided into smaller squares of equal size for reference. The first shape is a cube that is 3 one centimeter squares high and 3 across (for a total of nine centimeter squares cubes making up each side of the cube).

Calculation for surface area of the cube: 6 times (3 centimeters times 3 centimeters) = 54 centimeters squared.

Volume of the cube: 3 centimeters times 3 centimeters times 3 centimeters = 27 centimeters cubed.

The second shape is a flattened rectangle that is 3-one centimeter squares wide, one centimeter square in height and 9 centimeter squares long.

Calculation of the surface area in the rectangle: 2 times (3 centimeters times 1 centimeter) + 2 times (9 centimeters times 1 centimeter) + 2 times (3 centimeters times 9 centimeters) = 78 centimeters squared.

Volume: 1 centimeter times 3 centimeters times 9 centimeters = 27 centimeters cubed.

Thus, the cube has a surface area to volume ratio of 54 to 27, while the flattened rectangle has a surface area to volume ratio of 78 to 27. This demonstrates that the flattened shape has a much higher surface area relative to its volume than the fatter cube shape.

MAKE CONNECTIONS *Find other examples of flattening, folding, branching, and projections (see*Chapters 6*,***9**,**35**, and**42**). How is maximizing surface area important to the structure's function in each example?*

*For suggested answer, see*Appendix A.*

As you might expect since all their cells are close to water, flatworms have no organs specialized for gas exchange, and their relatively simple excretory apparatus functions mainly to maintain osmotic balance with their surroundings. This apparatus consists of **protonephridia**, networks of tubules with ciliated structures called *flame bulbs* that pull fluid through branched ducts opening to the outside (see Figure 44.8).

Most flatworms have a gastrovascular cavity with only one opening. Though flatworms lack a circulatory system, the fine branches of the gastrovascular cavity distribute food directly to the animal's cells.

Early in their evolutionary history, flatworms separated into two lineages, Catenulida and Rhabditophora. Catenulida is a small clade of about 100 flatworm species, most of which live in freshwater habitats. Catenulids typically reproduce asexually by budding at their posterior end. The offspring often produce their own buds before detaching from the parent, thereby forming a chain of two to four genetically identical individuals—hence their informal name, "chain worms."

The other ancient flatworm lineage, Rhabditophora, is a diverse clade of about 20,000 freshwater and marine species, one example of which is shown in Figure 33.8. We'll explore the rhabditophorans in more detail, focusing on free-living and parasitic members of this clade.

Free-Living Species

Free-living rhabditophorans are important as predators and scavengers in a wide range of freshwater and marine habitats. The best-known members of this group are freshwater species in the genus *Dugesia*, commonly called **planarians**. Abundant in unpolluted ponds and streams, planarians prey on smaller animals or feed on dead animals. They move by using cilia on their ventral surface, gliding along a film of mucus they secrete. Some other rhabditophorans also use their muscles to swim through water with an undulating motion.

A planarian's head features a pair of light-sensitive eyespots as well as lateral flaps that function mainly to detect specific chemicals. The planarian nervous system is more complex and centralized than the nerve nets of cnidarians **(Figure 33.9)**. Experiments have shown that planarians can learn to modify their responses to stimuli.

Figure 33.9

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Anatomy of a planarian.
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Planarian anatomy detailed, with the functions of pharynx, ganglia, ventral nerve cords and gastrovascular cavity. The planarian has a flat body with a roughly triangular anterior end containing two eyespots.

Digestion is completed within the cells lining the gastrovascular cavity, which has many fine subbranches that provide an extensive surface area. Undigested wastes are egested through an opening at the tip of the pharynx.

Ventral nerve cords. From the ganglia, a pair of ventral nerve cords runs the length of the body.

Pharynx. A muscular pharynx can be extended through the mouth. Digestive juices are spilled onto prey, and the pharynx sucks small pieces of food into the gastrovascular cavity, where digestion continues.

Ganglia. At the anterior end of the worm, near the main sources of sensory input, is a pair of ganglia, dense clusters of nerve cells. Body shape is flat. The anterior end is roughly triangular and contains two eyespots on the dorsal side as well as two nearby ganglia (one associated with each eyespot). The gastrovascular cavity lines the middle of the interior of the organism. Near the middle of the body, on the ventral side, is the mouth opening, through which the pharynx can extend (resembles a tube).

Some planarians can reproduce asexually through fission. The parent constricts roughly in the middle of its body, separating into a head end and a tail end; each end then regenerates the missing parts. Sexual reproduction also occurs. Planarians are hermaphrodites, and copulating mates typically cross-fertilize each other.

Parasitic Species

More than half of the known species of rhabditophorans live as parasites in or on other animals. Many have suckers that attach to the internal organs or outer surfaces of the host animal. In most species, a tough covering helps protect the parasites within their hosts. We'll discuss two ecologically and economically important subgroups of parasitic rhabditophorans, the trematodes and the tapeworms.

Trematodes

As a group, trematodes parasitize a wide range of hosts, and most species have complex life cycles with alternating sexual and asexual stages. Many trematodes require an intermediate host in which larvae develop before infecting the final host (usually a vertebrate), where the adult worms live. For example, various trematodes that parasitize humans spend part of their lives in snail hosts **(Figure 33.10)**. Around the world, about 200million people are infected with trematodes called blood flukes (*Schistosoma*) and suffer from schistosomiasis, a disease whose symptoms include pain, anemia, and diarrhea.

Figure 33.10

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    The life cycle of a blood fluke (*Schistosoma mansoni*), a
trematode.
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Lifecycle diagram depicts various stages and different hosts (human, snail) for a trematode.

The cycles is as follows.

1.Mature flukes live in the blood vessels of the human intestine (person drawn here pulling plants from shallow water). A female fluke fits into a groove running the length of the larger male's body (male appears wormlike in LM, scale 1 millimeter); female is a darker and smaller worm-like organism within the male.

2.Blood flukes reproduce sexually in the human host. The fertilized eggs exit the host in the feces.

3.If the human feces reach a pond or other source of water (as they do in this diagram) then the eggs develop into ciliated larvae (organism with cilia that appear as small hairs on the outer surface). These larvae infect snails, the intermediate host.

4.Asexual reproduction within a snail results in another type of motile larva, which escapes from the snail host. This larva is drawn as a longer than the larva in step 3, also with cilia and a branched tail.

5.These larvae penetrate the skin and blood vessels of humans working in fields irrigated with water contaminated with fluke larvae (and cycle can begin again at step 1).

WHAT IF? *Snails eat algae, whose growth is stimulated by nutrients found in fertilizer. How would the contamination of irrigation water with fertilizer likely affect the occurrence of schistosomiasis? Explain.*

*For suggested answer, see*Appendix A*.*

Living within more than one kind of host puts demands on trematodes that free-living animals don't face. A blood fluke, for instance, must evade the immune systems of both snails and humans. By mimicking the surface proteins of its hosts, the blood fluke creates a partial immunological camouflage for itself. It also releases molecules that manipulate the hosts' immune systems into tolerating the parasite's existence. These defenses are so effective that individual blood flukes can survive in humans for more than 40 years.

Tapeworms

The tapeworms are a second large and diverse group of parasitic rhabditophorans **(Figure 33.11)**. The adults live mostly inside vertebrates, including humans. In many tapeworms, the anterior end, or *scolex*, is armed with suckers and often hooks that the worm uses to attach itself to the intestinal lining of its host. Tapeworms lack a mouth and gastrovascular cavity; they simply absorb nutrients released by digestion in the host's intestine. Absorption occurs across the tapeworm's body surface.

Figure 33.11

Anatomy of a tapeworm.

Tapeworm body is long, ribbon-like and lined with proglottids with reproductive structures. The scolex, at one end, is seen in a close-up image; contains hooks and suckers. Scale: 100 micrometers.

The inset shows a close-up of the scolex (colorized SEM).

Posterior to the scolex is a long ribbon of units called *proglottids*, which are little more than sacs of sex organs. After sexual reproduction, proglottids loaded with thousands of fertilized eggs are released from the posterior end of a tapeworm and leave the host's body in feces. In one type of life cycle, feces carrying the eggs contaminate the food or water of intermediate hosts, such as pigs or cattle, and the tapeworm eggs develop into larvae that encyst in muscles of these animals. A human acquires the larvae by eating undercooked meat containing the cysts, and the worms develop into mature adults within the human. Large tapeworms can block the intestines and rob enough nutrients from the human host to cause nutritional deficiencies. Several different oral medications can kill the adult worms.

Rotifers and Acanthocephalans

Recent phylogenetic analyses have shown that two traditional animal phyla, the rotifers (former phylum Rotifera) and the acanthocephalans (former phylum Acanthocephala), should be combined into a single phylum, Syndermata. Each of the two groups has distinctive characteristics.

Rotifers

There are roughly 1,800 species of rotifers, tiny animals that inhabit freshwater, marine, and damp soil habitats. Ranging in size from about 50 μm to 2 mm, rotifers are smaller than many protists but nevertheless are multicellular and have specialized organ systems **(Figure 33.12)**. In contrast to cnidarians and flatworms, which have a gastrovascular cavity, rotifers have an **alimentary canal**, a digestive tube with two openings, a mouth and an anus. Internal organs lie within the *hemocoel* (see Figure 32.9b). Fluid in the hemocoel serves as a hydrostatic skeleton. Movement of a rotifer's body distributes the fluid throughout the body, circulating nutrients.

Figure 33.12

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A rotifer.
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Rotifer: Wine glass-shaped organism with crown of cilia around mouth with jaws behind, stomach in the center and anus at the base near what would be the wine glass stem. Scale: 0.1 millimeters.

Although smaller than many unicellular protists, rotifers are multicellular and have specialized organ systems (LM).

The word *rotifer* is derived from the Latin meaning “wheel-bearer,” a reference to the crown of cilia that draws a vortex of water into the mouth. Posterior to the mouth, rotifers have jaws called trophi that grind up food, mostly microorganisms suspended in the water. Digestion is then completed farther along the alimentary canal. Most other bilaterians also have an alimentary canal, which enables the stepwise digestion of a wide range of food particles.

Rotifers exhibit some unusual forms of reproduction. Some species consist only of females that produce more females from unfertilized eggs, a type of asexual reproduction called **parthenogenesis**. Some other invertebrates (for example, aphids and some bees) and even some vertebrates (for example, some lizards and some fishes) can also reproduce in this way. In addition to being able to produce females by parthenogenesis, some rotifers can also reproduce sexually under certain conditions, such as high levels of crowding. The resulting embryos can remain dormant for years. Once they break dormancy, the embryos develop into another generation of females that reproduce asexually.

It is puzzling that many rotifer species persist without males. The vast majority of animals and plants reproduce sexually at least some of the time, and sexual reproduction has certain advantages over asexual reproduction (see Concept 46.1). For example, species that reproduce asexually tend to accumulate harmful mutations in their genomes faster than sexually reproducing species. As a result, asexual species should experience higher rates of extinction.

Seeking to understand how they persist without males, researchers have been studying a clade of asexual rotifers named Bdelloidea. Some 360 species of bdelloid rotifers are known, and all of them reproduce by parthenogenesis without any males. Paleontologists have discovered bdelloid rotifers preserved in 35-million-year-old amber, and the morphology of these fossils resembles only the female form, with no evidence of males. Molecular clock analyses indicate that bdelloids have been asexual for over 50million years. While it appears that they do not reproduce sexually, bdelloid rotifers may be able to generate genetic diversity in other ways. For example, bdelloids can tolerate very high levels of desiccation. When conditions improve and their cells rehydrate, DNA from other species enters their cells through cracks in the plasma membrane. Recent evidence suggests that this foreign DNA can be incorporated into the bdelloids’ genome, thereby leading to increased genetic diversity.

Acanthocephalans

Acanthocephalans (1,100 species) are sexually reproducing parasites of vertebrates that lack a complete digestive tract and usually are less than 20cm long. They are commonly called spiny-headed worms because of the curved hooks on the proboscis at the anterior end of their body **(Figure 33.13)**. Acanthocephalans are triploblastic and, like rotifers, they have a hemocoel. Recent studies have shown that acanthocephalans originated from within the group traditionally known as Rotifera. In particular, rotifers in the genus *Seison*

share a more recent common ancestor with acanthocephalans than they do with other rotifers, making the acanthocephalans a group of highly modified “rotifers.”

Figure 33.13

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*Paratenuisentis ambiguus,* an acanthocephalan.
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Photo of Paratenuisentis ambiguus, shaped like a thin crescent, with close-up of the proboscis, which looks like a cactus with curved spines.

The inset photograph shows the curved hooks that give the spiny-headed worms their name.

All acanthocephalans are parasites that have complex life cycles with two or more hosts. Some species manipulate the behavior of their intermediate hosts (generally arthropods) in ways that increase their chances of reaching their final hosts (generally vertebrates). For example, acanthocephalans that infect New Zealand mud crabs cause their hosts to move to more visible areas on the beach, where the crabs are more likely to be eaten by birds, the worms’ final hosts.

Ectoprocts and Brachiopods

Bilaterians in the phyla Ectoprocta and Brachiopoda have a *lophophore*, a crown of ciliated tentacles around their mouth (see Figure 32.12a). As the cilia draw water toward the mouth, the tentacles trap suspended food particles. Other similarities, such as a U-shaped alimentary canal and the absence of a distinct head, reflect these organisms’ sessile existence. In contrast to flatworms, which lack a body cavity, and rotifers, which have a hemocoel, ectoprocts and brachiopods have a coelom (see Figure 32.9a).

Ectoprocts (from the Greek *ecto*, outside, and *procta*, anus) are colonial animals that superficially resemble clumps of moss. (In fact, their common name, bryozoans, means “moss animals.”) In most species, the colony is encased in a hard exoskeleton studded with pores through which the lophophores extend **(Figure 33.14a)**. Most ectoproct species live in the sea, where they are among the most widespread and numerous sessile animals. Several species are important reef builders. Ectoprocts also live in lakes and rivers. Colonies of the freshwater ectoproct *Pectinatella magnifica* grow on submerged sticks or rocks and can grow into a gelatinous, ball-shaped mass more than 10cm across.

Figure 33.14

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Ectoprocts and brachiopods.
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Two lophophorates.

Part a. Image of Ectoproct with stalk-like structure through which several fan-like lipophores extend.

Part b. Image of brachiopod with hinged shell through which fan-like lophophore extends. The two parts of the shell are dorsal and ventral.

Brachiopods, or lamp shells, superficially resemble clams and other hinge-shelled molluscs, but the two halves of the brachiopod shell are dorsal and ventral rather than lateral, as in clams **(Figure 33.14b)**. All brachiopods are marine. Most live attached to the seafloor by a stalk, opening their shell slightly to allow water to flow through the lophophore. The living brachiopods are remnants of a much richer past that included 30,000 species in the Paleozoic and Mesozoic eras. Some living brachiopods, such as those in the genus *Lingula*, appear nearly identical to fossils of species that lived 400millionyears ago.

Molluscs

Snails and slugs, oysters and clams, and octopuses and squids are all molluscs (phylum Mollusca). There are over 100,000 known species, making them the second most diverse phylum of animals (after the arthropods, discussed later). Although the majority of molluscs are marine, roughly 8,000 species inhabit fresh water, and 28,000 species of snails and slugs live on land. All molluscs are soft-bodied, and most secrete a hard protective shell made of calcium carbonate. Slugs, squids, and octopuses have a reduced internal shell or have lost their shell completely during their evolution.

Despite their apparent differences, all molluscs have a similar body plan **(Figure 33.15)**. The primary body cavity is a hemocoel, but they also have a reduced coelom. The body of a mollusc has three main parts: a muscular **foot**, usually used for movement; a **visceral mass** containing most of the internal organs; and a **mantle**, a fold of tissue that drapes over the visceral mass and secretes a shell (if one is present). In many molluscs, the mantle extends beyond the visceral mass, producing a water-filled chamber, the **mantle cavity**, which houses the gills, anus, and excretory pores. Many molluscs feed by using a straplike organ called a **radula** to scrape up food.

Figure 33.15

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The basic body plan of a mollusc.
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Diagram of a mollusc body plan shows a domed-shape with shell and two openings over the foot and many different organs, with some functions described.

Metanephridium. Excretory organs called metanephridia remove metabolic wastes from the hemolymph. Heart. Most mollusks have an open circulatory system. The dorsally located heart pumps circulatory fluid called hemolymph through arteries into sinuses (body spaces). The organs of the mollusc are thus continually bathed in hemolymph. The long digestive tract is coiled in the visceral mass.

Radula. The mouth region in many mollusc species contains a rasp-like feeding organ called a radula. This belt of backward-curved teeth repeatedly thrusts outward and then retracts into the mouth, scraping and scooping like a backhoe. The nervous system consists of a nerve ring around the esophagus, from which nerve cords extend. Basic body plan shown here is a dome-shaped shell, beneath which is the mantle and mantle cavity, followed by the visceral mass containing most of the organs. The foot is below (ventral to) the visceral mass and has branched nerve cords running through it, with one end of the nerve cord forming a ring around the esophagus. At the anterior end is the mouth region containing the radula, with a blade of backwards-curving teeth. At the posterior end is the anus. Other organs detailed in the visceral mass, from roughly anterior to posterior, are the esophagus (just posterior to the radula), and the stomach and intestines, which connect to the anus. Above (dorsal to) the stomach and intestines are the gonads followed by the coelom and the metanephridium and the heart, which is connected to the feather-like gill at the posterior of the body.

Most molluscs have separate sexes, and their gonads (ovaries or testes) are located in the visceral mass. Many snails, however, are hermaphrodites. The life cycle of many marine molluscs includes a ciliated larval stage, the trochophore (see **Figure 32.12b**), which is also characteristic of marine annelids (segmented worms) and some other lophotrochozoans.

The basic body plan of molluscs has evolved in various ways in the phylum's eight major clades. We'll examine four of those clades here: Polyplacophora (chitons), Gastropoda (snails and slugs), Bivalvia (clams, oysters, and other bivalves), and Cephalopoda (squids, octopuses, cuttlefishes, and chambered nautiluses). We will then focus on threats facing some groups of molluscs.

Chitons

Chitons have an oval-shaped body and a shell composed of eight dorsal plates **(Figure 33.16)**. The chiton's body itself, however, is unsegmented. You can find these marine animals clinging to rocks along the shore during low tide. If you try to dislodge a chiton by hand, you will be surprised at how well its foot, acting as a suction cup, grips the rock. A chiton can also use its foot to creep slowly over the rock surface. Chitons use their radula to scrape algae off the rock surface.

Figure 33.16

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A chiton.
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Image of chiton shows oval-shaped shell divided into 8 plate segments on a rock under water.

Note the eight-plate shell characteristic of molluscs in the clade Polyplacophora.

Gastropods

About three-quarters of all living species of molluscs are gastropods **(Figure 33.17)**. Most gastropods are marine, but there are also freshwater species. Still other gastropods have adapted to life on land, where snails and slugs thrive in habitats ranging from deserts to rain forests.

Figure 33.17

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Gastropods.
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Part a. Land snail has a cylindrical body with antenna on head and spiraling shell. Part b. Sea slug has a cylindrical body and bumpy-looking covering on dorsal side in place of shell.

The many species of gastropods have colonized terrestrial as well as aquatic environments.

Gastropods move literally at a snail's pace by a rippling motion of their foot or by means of cilia—a slow process that can leave them vulnerable to attack. Most gastropods have a single, spiraled shell into which the animal can retreat when threatened. The shell, which is secreted by glands at the edge of the mantle, has several functions, including protecting the animal's soft body from injury and dehydration. One of its most important roles is as a defense against predators, as is demonstrated by comparing populations with different histories of predation (see the **Scientific Skills Exercise**). As they move slowly about, most gastropods use their radula to graze on algae or plants. Several groups, however, are predators, and their radula has become modified for boring holes in the shells of other molluscs or for tearing apart prey. In the cone snails, the teeth of the radula act as poison darts that are used to subdue prey.

Scientific Skills Exercise

Understanding Experimental Design and Interpreting Data

AP Science Practices 3, 4, 6

Is There Evidence of Selection for Defensive Adaptations in Mollusc Populations Exposed to Predators? The fossil record shows that historically, increased risk to prey species from predators is often accompanied by increased incidence and expression of prey defenses. Researchers tested whether populations of the predatory European green crab (*Carcinus maenas*) have exerted similar selective pressures on its gastropod prey, the flat periwinkle (*Littorina obtusata*). Periwinkles from southern sites in the Gulf of Maine have experienced predation by European green crabs for over 100 generations, at about one generation per year. Periwinkles from northern sites in the Gulf have been interacting with the invasive green crabs for relatively few generations, as the invasive crabs spread to the northern Gulf comparatively recently.

A periwinkle is a small domed shell with a soft-bodied organism beneath.

A periwinkle

Previous research shows that (1) periwinkle shells recently collected from the Gulf are thicker than those collected in the late 1800s, and (2) periwinkle populations from southern sites have thicker shells than periwinkle populations from northern sites. In this exercise, you'll interpret the design and results of the researchers' experiment studying the rates of predation by crabs on periwinkles from northern and southern populations.

How the Experiment Was Done The researchers collected periwinkles and crabs from sites in the northern and southern Gulf of Maine, separated by 450 km of coastline. A single crab was placed in a cage with eight periwinkles of different sizes. After three days, researchers assessed the fate of the eight periwinkles. Four different treatments were set up, with crabs from northern or southern populations offered periwinkles from northern and southern populations. All crabs were of similar size and included equal numbers of males and females. Each experimental treatment was tested 12 to 14 times.

In a second part of the experiment, the bodies of periwinkles from northern and southern populations were removed from their shells and presented to crabs from northern and southern populations.

Data from the Experiment

Bar graph shows average numbers of northern and southern periwinkles killed by crabs. At least 5 times as many northern as southern periwinkles were killed regardless of crab source.

Data from R. Rochette et al., Interaction between an invasive decapod and a native gastropod: Predator foraging tactics and prey architectural defenses, *Marine Ecology Progress Series* 330:179–188 (2007).

One bar represents northern periwinkles and one represents southern periwinkles. For crabs from southern population sources, an average of a little more than 1 southern periwinkle was killed; for the same crab source population, a little more than 5 northern periwinkles were killed. For crabs from northern source

populations, average southern periwinkles killed were just above zero; for the same crab source population, an average of over 4 northern periwinkles were killed.

When the researchers presented the crabs with unshelled periwinkles, all the unshelled periwinkles were consumed in less than an hour.

INTERPRET THE DATA

- What hypotheses were the researchers testing in this study? What are the independent variables? What are the dependent variables?
- Why did the research team set up four different treatments?
- Why did researchers present unshelled periwinkles to the crabs? What do the results of this part of the experiment indicate?
- Summarize the results of the experiment. Do these results support the hypothesis you identified in question 1? Explain.
- Suggest how natural selection may have affected populations of flat periwinkles in the southern Gulf of Maine over the last 100 years.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Many gastropods have a head with eyes at the tips of tentacles. Terrestrial snails lack the gills typical of most aquatic gastropods. Instead, the lining of their mantle cavity functions as a lung, exchanging respiratory gases with the air.

Bivalves

The molluscs of the clade Bivalvia are all aquatic and include many species of clams, oysters, mussels, and scallops. Bivalves have a shell divided into two halves **(Figure 33.18)**. The halves are hinged, and powerful adductor muscles draw them tightly together to protect the animal's soft body. Bivalves have no distinct head, and the radula has been lost. Some bivalves have eyes and sensory tentacles along the outer edge of their mantle.

Figure 33.18

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A bivalve.
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A bivalve with two scallop-edged shells that hinge together, seen on a sand bed.

This scallop has many eyes (dark blue spots) lining each half of its hinged shell.

The gills of bivalves are used for feeding as well as gas exchange in most species **(Figure 33.19)**. Most bivalves are suspension feeders. They trap small food particles in mucus that coats their gills, and cilia then convey those particles to the mouth. Water enters the mantle cavity through an incurrent siphon, passes over the gills, and then exits the mantle cavity through an excurrent siphon.

Figure 33.19

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Anatomy of a clam.
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Organs and structures in a diagram of a clam are labeled.

Food particles suspended in water that enters through the incurrent siphon are collected by the gills and passed via cilia and the palps to the mouth.

Shell is roughly heart-shaped with a hinge at one end. Just inside the shell are the mantle and mantle cavity. Opposite the hinge area is the opening that is the incurrent siphon (water flows in); adjacent is the excurrent siphon (water flows out after passing over the gills). Near the excurrent siphon is the anus; adjacent are one or two adductor muscles. The anus is connected to the heart, gut, palp, digestive gland and mouth (opposite the anus). The foot is a large organ within the mantle cavity; also present are the gonad and coelom.

Most bivalves lead sedentary lives, a characteristic suited to suspension feeding. Mussels secrete strong threads that tether them to rocks, docks, boats, and the shells of other animals. However, clams can pull themselves into the sand or mud, using their muscular foot for an anchor, and scallops can skitter along the seafloor by flapping their shells, rather like the mechanical false teeth sold in novelty shops.

Cephalopods

Cephalopods are active marine predators **(Figure 33.20)**. They use their tentacles to grasp prey, which they then bite with beak-like jaws and immobilize with a poison present in their saliva. The foot of a cephalopod has become modified into a muscular excurrent siphon and part of the tentacles. Squids dart about by drawing water into their mantle cavity and then firing a jet of water through the excurrent siphon; they steer by pointing the siphon in different directions. Octopuses use a similar mechanism to escape predators.

Figure 33.20

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Cephalopods.
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Cephalopods. Squids are speedy carnivores with beak-like jaws and well-developed eyes. Octopuses are considered among the most intelligent invertebrates. Chambered nautiluses are the only living cephalopods with an external shell.

The mantle covers the visceral mass of cephalopods, but the shell is generally reduced and internal (in most species) or missing altogether (in some cuttlefishes and some octopuses). One small group of cephalopods with external shells, the chambered nautiluses, survives today.

Cephalopods are the only molluscs with a *closed circulatory system*, in which the blood remains separate from fluid in the body cavity. They also have well-developed sense organs and a complex brain. The ability to learn and behave in a complex manner is probably more critical to fast-moving predators than to sedentary animals such as clams.

The ancestors of octopuses and squids were probably shelled molluscs that took up a predatory lifestyle; the shell was lost in later evolution. Shelled cephalopods called **ammonites**, some of them as large as truck tires, were the dominant invertebrate predators of the seas for hundreds of millions of years until their disappearance during the mass extinction at the end of the Cretaceous period, 66 million years ago.

Most species of squid are less than 75 cm long, but some are much larger. The giant squid (*Architeuthis dux*), for example, has an estimated maximum length of 13 m for females and 10 m for males. The colossal squid (*Mesonychoteuthis hamiltoni*), is even larger, with an estimated maximum length of 14 m. Unlike *A. dux*, which has large suckers and small teeth on its tentacles, *M. hamiltoni* has two rows of sharp hooks at the ends of its tentacles that can inflict deadly lacerations.

It is likely that *A. dux* and *M. hamiltoni* spend most of their time in the deep ocean, where they may feed on large fishes. Remains of both giant squid species have been found in the stomachs of sperm whales, which are probably their only natural predator. Scientists first photographed *A. dux* in the wild in 2005 while it was attacking baited hooks at a depth of 900 m. *M. hamiltoni* has yet to be observed in nature. Overall, much remains to be learned about these marine giants.

Protecting Freshwater and Terrestrial Molluscs

Species extinction rates have increased dramatically in the last 400 years, raising concern that a sixth, human-caused mass extinction may be under way (see Concept 25.4). Among the many taxa under threat, molluscs have the dubious distinction of being the animal group with the largest number of documented extinctions **(Figure 33.21)**.

Figure 33.21

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The silent extinction.
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A pie chart is given, and next to that are a dozen workers on a mound of pearl mussels killed to make buttons, circa 1919.

Pie chart data are as follows:

Animal Group	Approximate proportion of pie chart
Molluscs	40 percent
Birds	21 percent
Fishes	18 percent
Mammals	14 percent
Insects	8 percent
Amphibians	6 percent
Reptiles (excluding birds)	3 percent
Other invertebrates	2 percent

Molluscs account for a largely unheralded but sobering 40% of all documented extinctions of animal species. These extinctions have resulted from human actions such as habitat loss and overharvesting. Many pearl mussel populations, for example, were driven to extinction by overharvesting for their shells, which were used to make buttons and other goods. Land snails are highly vulnerable to the same threats; like pearl mussels, they are among the world's most imperiled animal groups.

MAKE CONNECTIONS *Freshwater bivalves feed on and can reduce the abundance of photosynthetic protists and bacteria. As such, what types of effects would the extinction of freshwater bivalves likely have on aquatic communities (see*Concept 28.6*)?**

*For suggested answer, see*Appendix A*.**

Threats to molluscs are especially severe in two groups: freshwater bivalves and terrestrial gastropods. For example, the pearl mussels, a group of freshwater bivalves that can make natural pearls (gems that form when a mussel or oyster secretes layers of a lustrous coating around a grain of sand or other small irritant), are among the world's most endangered animals. Roughly 10% of the 300 pearl mussel species that once lived in North America have become extinct in the last 100 years, and over two-thirds of those that remain

are threatened by extinction. Terrestrial gastropods, such as the snail in Figure 33.21, are faring no better. Hundreds of Pacific island land snails have disappeared since 1800. Overall, more than 50% of the Pacific island land snails are extinct or under imminent threat of extinction.

Threats faced by freshwater and terrestrial molluscs include habitat loss, pollution, competition or predation by non-native species, and overharvesting by humans. Is it too late to protect these molluscs? In some locations, reducing water pollution and changing how water is released from dams have led to dramatic rebounds in pearl mussel populations. Such results provide hope that with corrective measures, other endangered mollusc species can be revived.

Annelids

Annelida means "little rings," referring to the annelid body's resemblance to a series of fused rings. Annelids are segmented worms that live in the sea, in most freshwater habitats, and in damp soil. Annelids, which have a coelom (and no hemocoel), range in length from less than 1mm to more than 3m.

Traditionally, the phylum Annelida was divided into three main groups, Polychaeta (the polychaetes), Oligochaeta (the oligochaetes), and Hirudinea (the leeches). The names of the first two of these groups reflected the relative number of chaetae, bristles made of chitin, on their bodies: polychaetes (from the Greek *poly*, many, and *chaitē*, long hair) have many more chaetae per segment than do oligochaetes.

However, a 2011 phylogenomic study and other recent molecular analyses have indicated that the oligochaetes are a subgroup of the polychaetes, making the polychaetes (as defined morphologically) a paraphyletic group. Likewise, the leeches have been shown to be a subgroup of the oligochaetes. As a result, these traditional names are no longer used to describe the evolutionary history of the annelids. Instead, current evidence indicates that the annelids can be divided into two major clades, Errantia and Sedentaria—a grouping that reflects broad differences in lifestyle.

Errantians

Clade Errantia (from the Old French *errant*, traveling) is a large and diverse group, most of whose members are marine **(Figure 33.22)**. As their name suggests, many errantians are mobile; some swim among the plankton (small, drifting organisms), while many others crawl on or burrow in the seafloor. Many are predators, while others are grazers that feed on large, multicellular algae. The group also includes some relatively immobile species, such as the tube-dwelling *Platynereis*, a marine species that recently has become a model organism for studying neurobiology and development.

Figure 33.22

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An errantian, the predator *Nereimyra punctata.*
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Photograph of Nereimyra punctata. Segmented, worm shaped body with parapodia on each segment (here, parapodia appear as long, fine, string-like structures).

This marine annelid ambushes prey from burrows it has constructed on the seafloor. N. punctata hunts by touch, detecting its prey with long sensory organs called cirri that extend from the burrow.

In many errantians, each body segment has a pair of prominent paddle-like or ridge-like structures called parapodia ("beside feet") that function in locomotion (see Figure 33.22). Each parapodium has numerous chaetae. (Possession of parapodia with numerous chaetae is not unique to Errantia, however, as some members of the other major clade of annelids, Sedentaria, also have these features.) In many species, the parapodia are richly supplied with blood vessels and also function as gills. Errantians also tend to have well-developed jaws and sensory organs, as might be expected of predators or grazers that move about in search of food.

Sedentarians

Species in the other major clade of annelids, Sedentaria (from the Latin *sedere*, sit), tend to be less mobile than those in Errantia. Some species burrow slowly through marine sediments or soil, while others live within tubes that protect and support their soft bodies. Tube-dwelling sedentarians often have elaborate gills or tentacles used for filter feeding **(Figure 33.23)**.

Figure 33.23

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The Christmas tree worm, *Spirobranchus giganteus.*
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Photograph of Christmas tree worm under water. Pictured are two central stalks, each with branchlike, feathery level structures that resemble bright red Christmas trees.

This sedentarian's two tree-shaped whorls are tentacles, which it uses in gas exchange and to collect food particles from the water. The tentacles emerge from a calcium carbonate tube secreted by the worm that protects and supports its soft body.

Although the Christmas tree worm shown in Figure 33.23 once was classified as a "polychaete," current evidence indicates it is a sedentarian. The clade Sedentaria also contains former "oligochaetes," including the two groups we turn to next, the leeches and the earthworms.

Leeches

Some leeches are parasites that suck blood by attaching temporarily to other animals, including humans **(Figure 33.24)**, but most are predators that feed on other invertebrates. Leeches range in length from 1 to 30cm. Most leeches inhabit fresh water, but there are also marine species and terrestrial leeches, which live in moist vegetation. Some parasitic species use bladelike jaws to slit the skin of their host. The host is usually oblivious to this attack because the leech secretes an anesthetic. After making the incision, the leech secretes a chemical, hirudin, which keeps the blood of the host from coagulating near the incision. The parasite then sucks as much blood as it can hold, often more than ten times its own weight. After this gorging, a leech can last for months without another meal.

Figure 33.24

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A leech.
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Photograph of a leech attached to a human thumb. Leech has dark, slick, segmented worm-like body and is only a little smaller than the thumb.

A nurse applied this medicinal leech (Hirudo medicinalis) to a patient's sore thumb to drain blood from a hematoma (an abnormal accumulation of blood around an internal injury).

Until the 20th century, leeches were frequently used for bloodletting. Today they are used to drain blood that accumulates in tissues following certain injuries or surgeries. In addition, forms of hirudin produced with recombinant DNA techniques can be used to dissolve unwanted blood clots that form during surgery or as a result of heart disease.

Earthworms

Earthworms eat their way through the soil, extracting nutrients as the soil passes through the alimentary canal. Undigested material, mixed with mucus secreted into the canal, is eliminated as fecal castings through the anus. Farmers value earthworms because the animals till and aerate the earth, and their castings improve the texture of the soil. (Charles Darwin estimated that one acre of farmland contains about 50,000 earthworms, producing 18tons of castings per year.)

A guided tour of the anatomy of an earthworm, which is representative of annelids, is shown in **Figure 33.25**. Earthworms are hermaphrodites, but they do cross-fertilize. Two earthworms mate by aligning themselves in opposite directions in such a way that they exchange sperm, and then they separate. Some earthworms can also reproduce asexually by fragmentation followed by regeneration.

Figure 33.25

Anatomy of an earthworm, a sedentarian.

Diagram of earthworm, with structures and functions labeled. Next to the diagram is a photograph of a man holding a giant Australian earthworm, which appears at least two feet long.

Each segment is surrounded by longitudinal muscle, which in turn is surrounded by circular muscle. Earthworms coordinate the contraction of these two sets of muscles to move. Parts are labeled as follows.

Coelom. The coelom of the earthworm is partitioned by septa.

Metanephridium. Each segment of the worm contains a pair of excretory tubules, called metanephridia, that discharge wastes from the blood and coelomic fluid through exterior pores. Tiny blood vessels are abundant in the earthworm's skin, which functions as its respiratory organ.

Ventral nerve cords. The nerve cords penetrate the septa and run the length of the animal, as do the digestive tract and longitudinal blood vessels.

Chaetae. Each segment has four pairs of chaetae, bristles that provide traction for burrowing. Many of the internal structures are repeated within each segment of the earthworm.

Cerebral ganglia. The earthworm nervous system features a brainlike pair of cerebral ganglia above and in front of the pharynx. A ring of nerves around the pharynx connects to a subpharyngeal ganglion, from which a fused pair of nerve cords runs posteriorly. The circulatory system, a network of vessels, is closed. The dorsal and ventral vessels are linked by segmental pairs of vessels, some of which are muscular and pump blood through the circulatory system.

Diagram shows earthworm from mouth end towards anus end, detailing structures in between. There is also a cutaway cross-section of a segment of the earthworm labeled. Mouth is connected to tube that runs length of earthworm (ending at anus). Behind mouth is subpharyngeal ganglion attached to the ventral nerve cords (which run parallel to the alimentary canal) and the cerebral ganglia. Behind the mouth on the alimentary canal are the pharynx and then the esophagus. Blood vessels loop around the alimentary canal. Posterior to the esophagus is the crop, which is a bulge in the alimentary canal, and then the gizzard, followed by the intestine. Metanephridium are drawn adjacent to the alimentary canal. On the outer surface of the worm are tiny hair-like structures called chaetae; the clitellum is also on diagrammed as a bulge on the outer surface of the worm, closer to the head than the anus. The cross-section of the worm segment shows the longitudinal and circular muscles beneath the epidermis (skin), and there is a cuticle on the outer surface of the worm. The septum is a partition between segments. The alimentary canal is in the center of the worm's body and the dorsal vessel runs parallel. Fused nerve cords, nephrostomes, and metanephridium are also labeled.

As a group, Lophotrochozoa encompasses a remarkable range of body plans, as illustrated by members of such phyla as Syndermata, Ectoprocta, Mollusca, and Annelida. Next we'll explore the diversity of Ecdysozoa, a dominant presence on Earth in terms of sheer number of species.